ODE-FFN: Progress Report 2

Independent Study — Deep Learning

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3 November 2026 (Draft)

1 Project and main progress

This project asks whether ODE-inspired FFNs improve JiT's feature processing. The original design composes two Taylor maps around SiLU. We followed the professor's suggestions to study **Muon and matrix spectra** and **alternative input diagonals** $PD(h)$. Muon improves some original-layer fits while P grows. A bounded diagonal is competitive on Ring8, but ordinary ISRU explains much of its fitting benefit. No general ODE-specific gain is established.

The new studies use a width-64, three-block, time-conditioned residual MLP, **without JiT attention or image patches**. Noise–data pairs train velocity from $z_t = tX+(1-t)e$. Ring8 and Product16 mixtures provide analytic velocities. New comparisons use 20k updates, batch 512 and Heun32. Validation oracle MSE measures fitting; separate test SW2 measures generation. Older results retain their stated budgets.

For the original scalar layer at $K = 2$,

$$T(h) = h + s(h)Ph + \tfrac{1}{2}s(h)^2P^2h, \qquad s(h) = 0.2 + 0.8\,\sigma(g(h)/10). \tag{1}$$

The internal clock s differs from FM time t. Its input derivative contributes to the full layer Jacobian. The factorial matrix series converges for every finite P; a spectral radius below one is not required.

2 Professor suggestion 1: Muon and the spectrum

Purpose. Test whether Muon's matrix-update geometry improves fitting and controls P. Muon approximately orthogonalizes FFN momentum updates; other parameters retain AdamW. It does not directly constrain P.

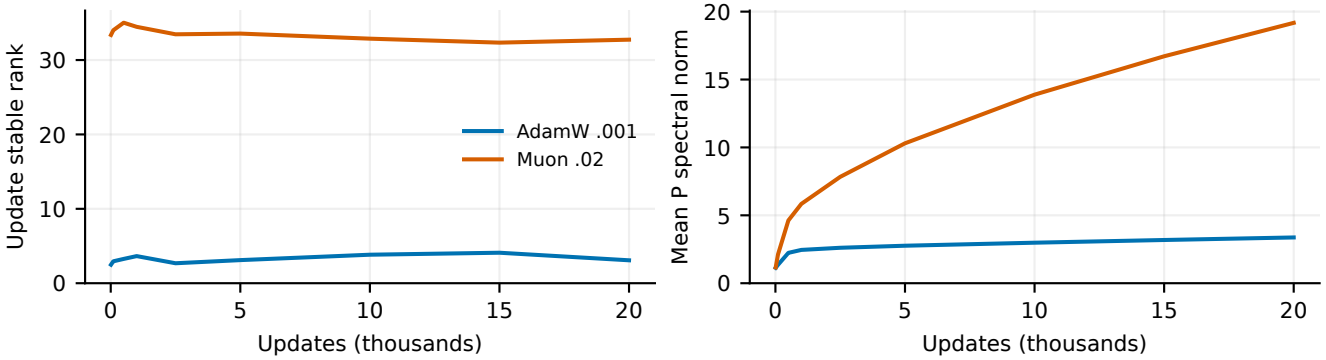


Figure 1: Product16 scalar K2, six-matrix means. Matrix rates: AdamW .001 and Muon .02. Update stable rank is $\|\Delta P\|_F^2/\|\Delta P\|_2^2$. A broader update spectrum coexists with larger P.

Observation. Update stable rank rises from 3.08 to 32.74, while mean P spectral norm rises from 3.37 to 19.17. Field MSE improves from 0.03343 to 0.02385; SwiGLU192 reaches 0.01567. Norm-matched momentum does not reproduce all Muon gains, while tested matrix decay harms fitting.

Constraint control. Flattening P at preserved Frobenius energy, paired with operator-norm-matched contraction, worsened both metrics during training. Frozen flattening also failed as a general repair.

3 Professor suggestion 2: alternative designs for $PD(h)$

Purpose. Replace the scalar clock by channel weights $d(h)$ and separate controller design, nonlinear features and order. With $A(h) = P\,\mathrm{diag}(d(h))$,

$$T_2(h) = h + A(h)h + \tfrac{1}{2}A(h)^2h, \tag{2}$$
$$F_1(h) = h + P[d(h) \odot h], \quad J_{F_1} = I + P\big[\mathrm{diag}(d) + \mathrm{diag}(h)J_d\big]. \tag{3}$$

The Jacobian contains feedback through $d(h)$, omitted by conditional spectra. For a free affine controller, $P' = PC^{-1}$ and $D' = CD$ preserve PD while changing P's spectrum, for invertible constant diagonal C. Bounded gates need not permit this reparameterization.

Design question	Controlled test and observation
Learned channel weights	MLP and rank-32 linear controllers, crossed with sigmoid/affine output. Some weak-parent gains; none beats the near-parameter SwiGLU control. Frozen output swaps do not repair the gap.
Simple input diagonals	Constant, raw h, sigmoid, shifted $h+1$ and $\lvert h\rvert$. Raw K1 gives a small Ring8 field lead with worse sampling; absolute input worsens its matched fits.
Polynomial order	Scalar K1 improves field fitting in every matched K1/K2 setting. Raw K1 at 80k still loses sampling or wider ordinary controls.
Growth control	Radial $d = h/\sqrt{1+\mathrm{mean}(h^2)}$ versus raw and fixed scaling. Hidden responses shrink substantially without a qualifying quality gain.
Bounded odd feature	$d_i = (1+1.2h_i^2)^{-1/2}$, compared with constant D, ordinary free ISRU and three SwiGLU widths; results below.

Table 1: Each row isolates a question; the evidence index retains every setting.

Bounded odd feature. The earlier ordinary ISRU lead motivates $q_i = h_i/\sqrt{1+1.2h_i^2}$ inside the diagonal construction. Each K1 solve is $h + Pq(h)$, keeping both solves and middle SiLU. Since $q_i' = (1+1.2h_i^2)^{-3/2}$, $\|J_{F_1}\|_2 \le 1 + \|P\|_2$; constant D has the same bound. Ordinary free ISRU, $W_2q(W_1h)$, retains its historical initializer and independently trained maps.

		Ring8		Product16	
FFN	Optimizer	MSE	SW2	MSE	SW2
Constant D	A	0.01441	0.19880	0.07541	0.10158
Constant D	M	0.01234	0.17134	0.01896	**0.07552**
ISRU-D	A	**0.00977**	**0.14591**	0.01606	0.08574
ISRU-D	M	0.01293	0.16713	0.02814	0.11673
Free ISRU	A	0.01000	0.17339	**0.01536**	0.08432
Free ISRU	M	0.02273	0.14728	0.02742	0.09024
SwiGLU64	selected	0.01128	0.15038	0.01878	0.08106
SwiGLU128	selected	0.01148	0.16452	0.01685	0.08185
SwiGLU192	selected	0.01123	0.16580	0.01567	0.07921

Table 2: Complete new 20k comparison; lower is better. A: AdamW .0003; M: Muon .02. Selected SwiGLU uses Muon .02 on Ring8 and AdamW .0003 on Product16, except Swi64 at .001. Auxiliary parameters always use AdamW .0003. ISRU-D/free/constant have 58,512 Product16 parameters versus Swi64's 58,572.

Constant D shares every initial tensor with ISRU-D. Free ISRU changes the FFN parameterization and initial function. Baseline states and generated arrays reproduce exactly.

4 What these results change

A positive result with an ordinary control. AdamW ISRU-D improves Ring8 field fitting and SW2 over all three SwiGLU widths. Its field lead over free ISRU is only 2.25%, below the registered 5% continuation margin, although its sampling distance is lower. On Product16, free ISRU and SwiGLU192 fit better. The large gain over constant-D AdamW mainly corrects a weak parent; it does not establish an ODE-specific benefit. The constant-D Muon row gives the lowest Product16 SW2 despite worse field fitting.

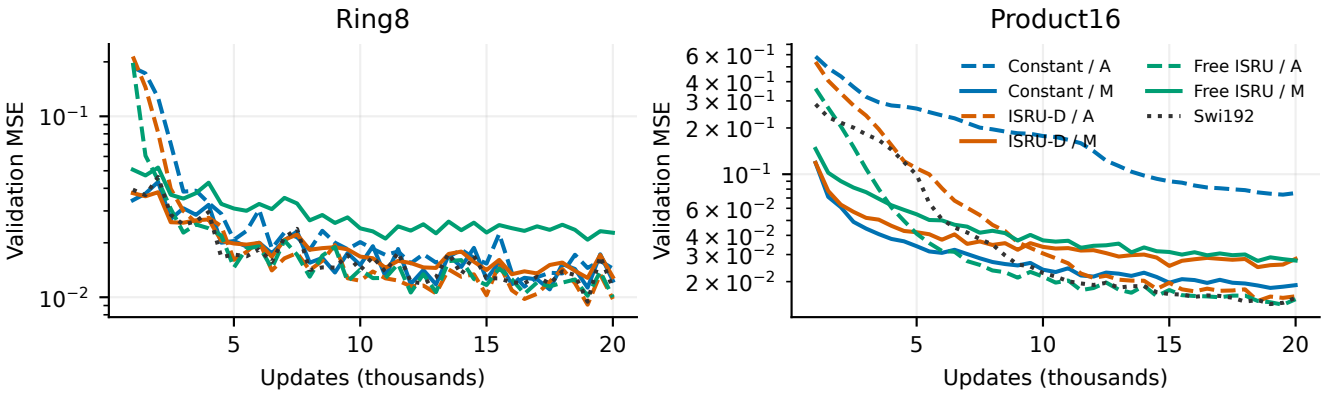


Figure 2: Diagonal training curves at Table 2 settings; dotted: SwiGLU192. Muon helps constant D but worsens both ISRU fits here. Comparisons use the fixed 20k endpoint; convergence is not established.

Earlier positive cases remain relevant. At the earlier 5k budget, a complete-gradient field readout lowers original-ODE Product16 MSE from 0.325527 to 0.037320, while an ordinary gradient SwiGLU reaches 0.033826. Input-dependent orthogonal frames give thin-target MSE 0.000685 at K2 and 0.000713 at K1, versus SwiGLU192's 0.003095. Their pointwise spectra are unchanged by conjugation, so useful adaptation can occur outside raw spectral shape. However, on the thick target, frame K2 tangent SW2 is 0.3173 versus SwiGLU192's 0.2471. These gains concern learned computation and directions, with distinct protocols that cannot be ranked against the new 20k results.

Limits. For $q = D(h)h$, increment energy is $\mathbb{E}\|Pq\|^2 = \mathrm{tr}(PMP^\top)$, $M = \mathbb{E}[qq^\top]$. Feature statistics matter alongside P. ISRU is an ordinary response, and the retained SiLU means the complete ISRU-D FFN is not odd. The evidence does not establish a need for K2 or transfer to JiT. Image comparisons still require an ordinary baseline below the project's FID-50 gate. The full attempt ledger and current evidence index retain earlier positive cases, failed settings and source identities.

5 Hypotheses for the next two weeks

1. **Muon helps through the feature response, rather than raw P regularity.** At saved states, compare native and norm-matched updates on the same minibatch. Measure changes in Pq and full-field loss alongside spectra. Test which quantities predict one-step improvement before selecting another constraint; separately trained trajectories cannot establish causation.
2. **The Ring8 sampling lead reflects where field errors occur during transport.** Compare saved ISRU-D, free-ISRU and ordinary-control checkpoints against the analytic field at common times and generated-path states. Fix checkpoints and solver resolution to distinguish error location from average MSE before proposing another architecture.